

Same Start, Different Weight Spaces: How AdamW and Muon Adapt Dense Retrievers

Anonymous ACL submission

SYNTHETIC LAYOUT TEST – NOT EXPERIMENT RESULTS

Abstract

Changing the optimizer changes the weight state that serves retrieval. We compare AdamW, Muon, and NorMuon from one pretrained DenseOn checkpoint on 500,000 shared examples; five checkpoints from 12 configurations are evaluated on 14 decontaminated BEIR tasks. Across the four-rate surface, Muon versus AdamW is positive at +0.0064 [+0.0056, +0.0071] nDCG@10, and NorMuon versus AdamW is positive at +0.0154 [+0.0131, +0.0178]. We trace hidden-weight displacement, subspace change, and functional use of 768 dimensions; candidate mechanisms must predict retrieval at unseen learning-rate doses. Because Muon’s update spectrum is induced by its definition, we treat it as an operator fingerprint rather than an explanation. We then test how two fixed reached states respond to reset continuation operators. At the 50K continuation endpoint, state, operator, and interaction contrasts are +0.0200 [+0.0100, +0.0300], +0.0200 [+0.0100, +0.0300], and +0.0200 [+0.0100, +0.0300] nDCG@10. They do not decompose the primary-training contrast.

1 Introduction

Dense retrievers are usually adapted with AdamW. This convention hides a scientific choice: starting from the same pretrained checkpoint, an optimizer selects a path through weight space and therefore the representation used for corpus retrieval. Muon makes a qualitatively different choice. It approximately orthogonalizes momentum updates for hidden matrices and remains trainable at numerical learning rates far above the AdamW range (Loshchilov and Hutter, 2019; Liu et al., 2025). The visible consequence—a flatter update spectrum—is largely built into the rule. Our question is not whether that signature appears, but

what weight state repeated Muon updates create and why that state retrieves differently.

This distinction matters because an optimizer update is not a retrieval outcome. Each accepted step changes the weights, future gradients, and optimizer state on which later steps act. Two rules can therefore have recognizable local geometry yet reach states with similar functions; conversely, they can have similar immediate effects and diverge after a long closed-loop trajectory. A useful mechanism must connect

updates $\rightarrow$ weights $\rightarrow$ embeddings $\rightarrow$ retrieval.

We organize evidence around these links, keeping predictive associations separate from intervention contrasts rather than equating the first link with the last.

We adapt DenseOn from one common initialization on a deterministic 500,000-query dataset with one positive and seven hard negatives per query. AdamW, Muon, and NorMuon each receive four predeclared learning rates and five checkpoint evaluations. Our primary comparison averages the complete rate surface within optimizer over 14 decontaminated BEIR tasks; the full trajectories show when differences emerge. We then measure hidden-weight displacement and subspace change, and ask a more functional question inspired by coordinate ablation: does an optimizer change how broadly and constructively the 768-dimensional embedding space contributes to retrieval?

Figure 1 summarizes the identification strategy. Descriptive geometry is not promoted to a mechanism merely because it separates optimizers. A candidate must improve retrieval prediction when an entire learning-rate dose is held out. The 2×2 continuation study crosses two fixed reached checkpoints with reset AdamW or Muon, calibrating hidden update directions on a common probe. It tests whether the relative continuation response depends

on the reached weights, not whether the original training gain can be partitioned into three mechanisms.

This investigation makes three contributions:

- a complete optimizer–rate–time surface for DenseOn adaptation, evaluated by full-corpus retrieval rather than a single selected checkpoint;
- a trajectory analysis that connects hidden-weight motion to representation rank, random dimension removal, and helpful versus degrading coordinate contributions, with a shared-rotation control and out-of-dose prediction; and
- a crossed reset continuation that tests state-dependent responses to further optimization, without treating an operator fingerprint as a causal explanation of the original gain.

2 From Optimizer Updates to Retrieval States

2.1 The loss sees a shortlist; evaluation sees a corpus

For query q_i , positive d_i^+ , and seven sampled negatives C_i^- , let $z_{id} = s(q_i, d)/\tau$. The contrastive loss is

$$\ell_i = -z_i^+ + \log Z_i, \quad (1)$$

$$Z_i = \exp(z_i^+) + \sum_{d \in C_i^-} \exp(z_{id}). \quad (2)$$

Full-corpus retrieval instead depends on the rank of d_i^+ among all documents in $\mathcal{D}_i$. Holding out queries prevents direct query memorization, but it does not make C_i^- representative of $\mathcal{D}_i \setminus C_i^-$. Consequently, validation loss can improve while nDCG falls. We call a change in optimizer ordering between these measurements the *shortlist–corpus gap*.

2.2 Update geometry is not an outcome

For a matrix parameter W with gradient G_t , AdamW applies an element-wise second-moment preconditioner. Muon instead applies a Newton–Schulz approximation to the polar factor of a momentum matrix. If $M = U\Sigma V^\top$, its idealized transform is $UV^\top$, which approximately equalizes nonzero singular values (Jordan et al., 2024). Nor-Muon adds a row-wise historical rescaling after orthogonalization (Li et al., 2025).

These are operator properties, not retrieval explanations. A flatter update can amplify weak or noisy directions, and large weight motion can leave the ranking nearly unchanged (Pang et al., 2026; Qu et al., 2026). We therefore study saved weight states and their representations along complete trajectories. Coordinate interventions and out-of-dose prediction test functional consequences of those states; crossed continuation tests their response to further optimization. None of these is a direct intervention on a measured weight-geometry feature, so predictive support is not causal mediation.

3 Related Work

Dense dual encoders compress queries and documents into single vectors and rank by similarity. BEIR evaluates whether such representations transfer across heterogeneous retrieval tasks (Thakur et al., 2021). DenseOn provides the unsupervised, not-yet-fine-tuned checkpoint and open training recipe used here (Sourty et al., 2026). Work on representation isotropy and effective rank motivates possible bridges between optimization and retrieval, but those statistics are not surrogates for ranking quality (Kim et al., 2024).

Coordinate ablations provide a more functional view of dimension use. Takeshita et al. (2025) find that text encoders often retain most retrieval quality after random removal of half their dimensions; their tested isotropy, correlation, and outlier measures do not account for the observation, while deleting some individual coordinates improves performance. We use this distinction to ask whether an optimizer merely changes representation rank or changes the distribution of helpful, redundant, and harmful dimensions along its trajectory.

Muon has shown attractive scaling in language-model pretraining, while NorMuon augments its matrix-level transform with neuron-wise adaptation (Liu et al., 2025; Li et al., 2025). Concurrent work warns that switching an Adam-pretrained model to Muon can disrupt pretrained knowledge as update strength grows (Qu et al., 2026; Liu et al., 2026). Other analyses relate Muon to feature rank or margins, identify risks from spectral equalization, or argue for covariance-aware and layer-dependent matrix geometry (Ruan et al., 2026; Pang et al., 2026; Du and Su, 2026; Massena et al., 2026). Our question is downstream and retrieval-specific: which geometric changes survive optimizer state feedback, preserve unseen scores, and predict cor-

From optimizer trajectories to retrieval function

Shared start and training data; compare reached weights, functional dimensions, and retrieval.

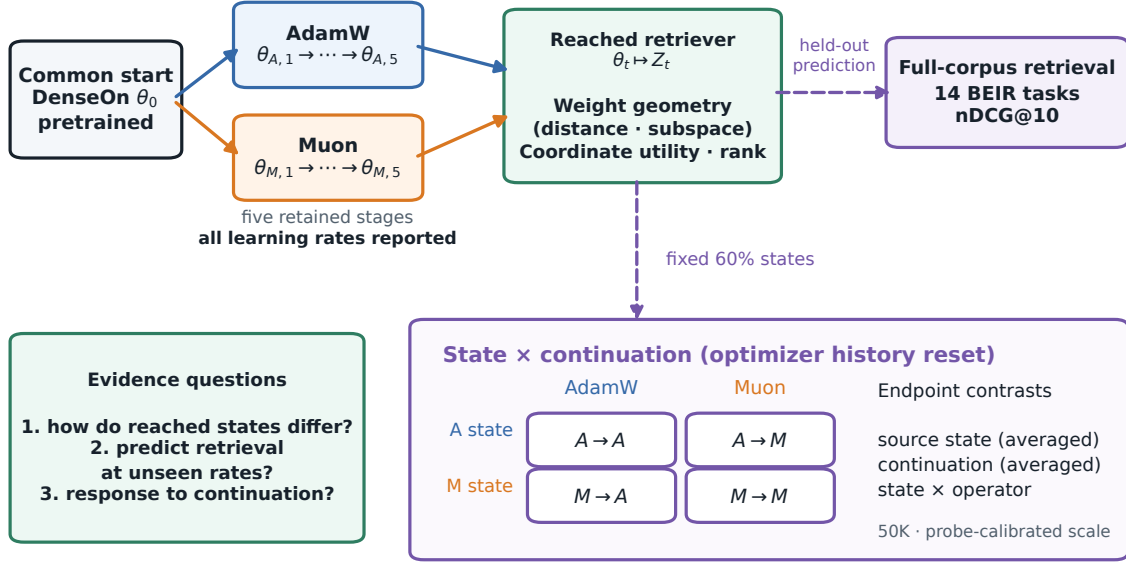


Figure 1: Our explanatory target and evidence chain. Optimizer-specific trajectories create reached weight states and embeddings; weight and coordinate-level measurements become candidate explanations only if they predict full-corpus retrieval out of learning-rate dose. The crossed continuation estimates post-continuation state, operator, and interaction contrasts for one fixed checkpoint pair.

pus ranking?

4 Tracing an Optimizer Through Weight Space

4.1 One controlled adaptation problem

We adapt the released DenseOn unsupervised checkpoint on one deterministic view of 500,000 queries. Each group contains one positive and 7 distinct sampled hard negatives. Queries are allocated proportionally across seven source datasets; a candidate is eligible only when its mined score is below 0.95 times the positive score, and the seven negatives are sampled without replacement from the first ten eligible candidates using stable seed-42 hashes. Every optimizer receives identical group identities and order. We use cosine InfoNCE at temperature 0.02, no in-batch or cross-device negatives, maximum context length 8,192, global batch size 128, one epoch, and bfloat16 forward passes.

The grid contains four learning rates per optimizer: $\{10^{-6}, 3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}\}$ for AdamW and $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ for Muon and NorMuon. Muon-family runs route hidden two-dimensional matrices through the matrix rule and auxiliary parameters through AdamW at 3×10^{-6} .

Each run keeps checkpoints at 20, 40, 60, 80, and 100% of 3,907 optimizer steps. Every checkpoint is evaluated on the same 14-task decontaminated BEIR suite.

We use eight NVIDIA L20Z GPUs as two disjoint four-GPU training pools, retaining four ranks per run. Retrieval evaluation is task-parallel across otherwise idle devices and uses exact retrieval rather than approximate ranking. AdamW uses fused CUDA AdamW; Muon and NorMuon use the documented decomposed-bfloat16 Newton-Schulz implementation. All runs retain the same example identities, ordering, checkpoint boundaries, and scoring semantics.

4.2 Evidence and inference

The primary estimand averages the four predeclared learning-rate cells within each optimizer at the final checkpoint. This makes the optimizer comparison a statement about the tested rate surface, not a post-hoc best cell. Task-level common resampling supplies simultaneous max- T intervals for the three pairwise contrasts. We also report all five optimizer-by-stage means and a secondary validation-selected comparison, but neither can re-

Contrast	Mean	95% CI	Decision
Muon - AdamW	+0.0064	[+0.0056, +0.0071]	positive
NorMuon - AdamW	+0.0154	[+0.0131, +0.0178]	positive
NorMuon - Muon	+0.0091	[+0.0075, +0.0107]	positive

Table 1: Final-stage retrieval over 14 decontaminated BEIR tasks, with simultaneous paired-task intervals for all three optimizer contrasts.

place the primary surface-level result.

Mechanism evidence is deliberately harder to obtain than optimizer separation. Nine frozen weight-space features are added one at a time to a baseline containing optimizer, stage, and within-optimizer learning-rate dose. We evaluate each feature by leaving one dose index out across all optimizers. A feature is supported only if it lowers pooled held-out RMSE and improves at least three of four folds. Coordinate-level representation features follow the same out-of-dose rule. Descriptive correlation, optimizer classification, and visually separated trajectories cannot pass this gate.

5 Optimizer Effects on Dense Retrieval

The primary estimand averages all four declared rates within optimizer from a common model, dataset, training schedule, and evaluation suite.

Primary comparison. Across the four-rate surface, Muon versus AdamW is positive at +0.0064 [+0.0056, +0.0071] nDCG@10, and NorMuon versus AdamW is positive at +0.0154 [+0.0131, +0.0178]. Validation-selected comparisons are secondary: Muon versus AdamW positive at +0.0090 [+0.0081, +0.0100]; NorMuon versus AdamW positive at +0.0185 [+0.0158, +0.0211]; NorMuon versus Muon positive at +0.0094 [+0.0077, +0.0111]. These are one-seed, pinned-grid results; learning rates are not independent training seeds. An inconclusive interval is not evidence of equivalence.

6 How the Optimizers Reshape the Retriever

6.1 Weight-state trajectories

For every hidden matrix W_ℓ , we measure the saved-checkpoint segment $W_{\ell,t} - W_{\ell,t-1}$ and cumulative displacement $W_{\ell,t} - W_{\ell,0}$ relative to the current weight norm. Randomized sketches estimate stable and entropy effective rank, and rank-16 left/right subspace overlap compares all unordered run pairs at each stage. Optimizer-pair summaries average all learning-rate pairs equally, without selecting a

reference rate. These are measurements of the state produced by many updates; we do not relabel a retained-checkpoint segment as a single optimizer step.

At the final stage, all-rate mean cumulative hidden-weight displacement relative to weight norm is AdamW 0.217725, Muon 0.426456, NorMuon 0.540762; the corresponding last-segment ratios are AdamW 0.108831, Muon 0.213417, NorMuon 0.270603.

6.2 Does Muon use embedding dimensions differently?

Weight motion matters only through the representation it creates. On a frozen 224-query probe, we therefore measure query/document covariance rank and recompute retrieval after removing shared random coordinate sets at 10, 25, 50, and 75%. For coordinate j , let $a_j = m(Z_{-j}) - m(Z)$, where m is shortlist nDCG or positive-hardest-negative margin after renormalization. Positive a_j marks a degrading coordinate; negative a_j marks a helpful one. We summarize helpful mass, degrading mass, and the participation ratio of helpful attribution rather than treating effective rank alone as dimension use. These leave-one-coordinate-out sensitivities are not an additive decomposition of the score: renormalization and changes in document ranking couple their effects.

Coordinate attribution is basis dependent. We repeat the endpoint analysis after three shared Haar rotations, each of which preserves all full-vector cosine scores. A native-coordinate difference can support a claim about native-coordinate utilization. Directional agreement across the three tested rotations is a robustness check, not proof of invariance under arbitrary bases or of greater dimensional capacity.

Muon does not meet the joint native-coordinate criterion relative to AdamW; this does not establish equivalence. The required mean directions do not hold under every tested shared rotation. NorMuon does not meet the joint native-coordinate criterion relative to AdamW; this does not establish equivalence. The required mean directions do not hold under every tested shared rotation. With 50% of coordinates removed, relative shortlist nDCG is 1.01449/1.01443/1.00878 for AdamW/Muon/NorMuon. The figure reports all declared rates and stages.

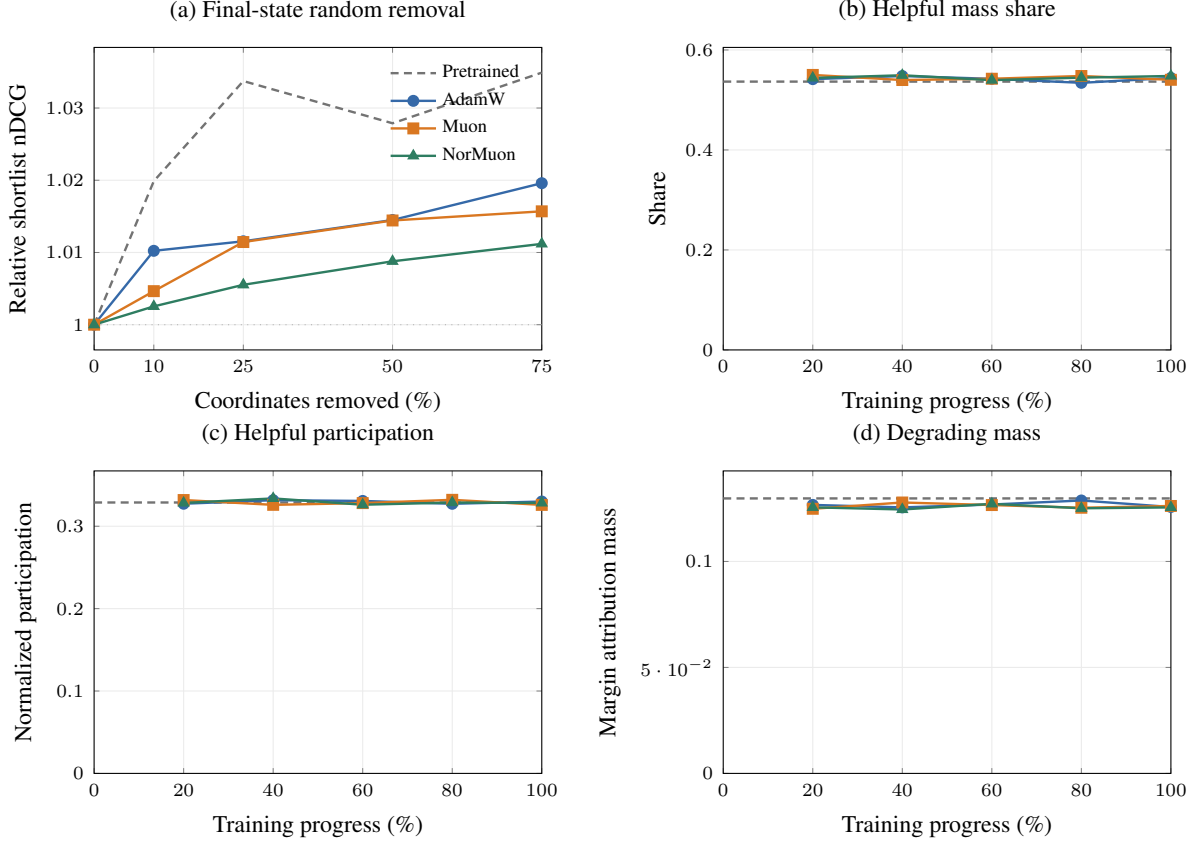


Figure 2: Dimension utility over the complete rate grid. (a) Final-state retention after shared random coordinate removal; the zero-removal value is one by definition. (b–d) Native-coordinate margin attribution at all five stages. Each optimizer curve averages all four rates and 14 tasks; (a) additionally averages 20 shared masks per nonzero fraction. The gray curve or horizontal line is the pretrained reference. Lines join measured points, not fitted trajectories. These are descriptive means, not confidence intervals or full-corpus retrieval scores; formal contrasts and rotation controls are reported separately.

6.3 Which state measurements predict retrieval?

Every weight- and representation-space candidate faces the same leave-dose-index-out test. The baseline already knows optimizer identity, checkpoint stage, and within-optimizer dose, so a feature must explain variation beyond recognizing Muon. We report all candidates, including unsupported ones, and treat predictive support as a bridge to retrieval rather than causal mediation.

The declared out-of-dose predictive criterion supports: log segment/weight norm, cumulative/weight norm, cumulative stable-rank fraction. All nine features and all four folds are retained. Undefined folds are not pooled as available cases. This is prediction outside a learning-rate dose on the same task suite, not causal mediation, statistical significance or evidence of equivalence.

Helpful mass share does not meet the declared held-out-dose predictive criterion. Helpful par-

ticipation does not meet the declared held-out-dose predictive criterion. Degrading mass does not meet the declared held-out-dose predictive criterion. 50% random-removal retention does not meet the declared held-out-dose predictive criterion. This tests prediction outside a learning-rate dose on the same task suite, not causal mediation or generalization to unseen tasks.

7 How Do Reached States Respond to Further Optimization?

Retrieval at a fixed checkpoint depends on its weights, not on an optimizer applied only later. Our follow-up therefore asks a conditional question: does the reached state change the response to further optimization? We fix the 60% AdamW 3×10^{-5} and Muon 3×10^{-4} checkpoints, reset optimizer history, and continue each with reset AdamW and reset Muon. The source pair is fixed, not selected from the new BEIR outcomes or averaged

over the primary rate surface.

Each cell’s hidden learning rate is calibrated to a global hidden update-to-weight Frobenius ratio of 5×10^{-4} on the fixed 32-example gradient-history probe, then held fixed as the schedule’s base rate. This does not match the first realized shuffled-batch update or subsequent update norms. Both rules use the same hidden/auxiliary partition and auxiliary AdamW at 3×10^{-6} ; calibration excludes weight decay. The four cells share each of three fixed data orders and a 50,000-query horizon.

For each seed–task pair, let Y_{so} be final nDCG@10 after this continuation, with source $s \in \{A, M\}$ and continuation rule $o \in \{A, M\}$. The frozen contrasts are

$$\Delta_S = \frac{1}{2}[(Y_{MA} - Y_{AA}) + (Y_{MM} - Y_{AM})], \quad (3)$$

$$\Delta_O = \frac{1}{2}[(Y_{AM} - Y_{AA}) + (Y_{MM} - Y_{MA})], \quad (4)$$

$$\Delta_I = (Y_{MM} - Y_{MA}) - (Y_{AM} - Y_{AA}). \quad (5)$$

They concern post-continuation endpoints, not gains relative to each source checkpoint. In particular, $Y_{MM} - Y_{AA} = \Delta_S + \Delta_O$; the interaction is not a third additive component. A positive averaged contrast need not hold within both levels, and positive interaction can mean less harm rather than improvement. Each contrast receives its own 95% two-way seed/task bootstrap interval from 100,000 resamples; these are marginal, not simultaneous intervals. An inconclusive contrast is not equivalent to zero or dominated by one whose interval excludes zero.

Averaged over continuation operators, the Muon-reached versus AdamW-reached state changes final nDCG@10 by +0.0200 [+0.0100, +0.0300] (positive); averaged over source states, Muon versus AdamW continuation changes it by +0.0200 [+0.0100, +0.0300] (positive). The state–operator interaction is +0.0200 [+0.0100, +0.0300] (positive). The relative Muon-versus-AdamW continuation effect is more favorable from the Muon-reached state at the tested horizon; this need not mean Muon helps from either state alongside a positive averaged source-state contrast and a positive averaged continuation contrast. This interaction does not establish full-trajectory co-adaptation.

8 Discussion

Optimizer quality is a property of a trajectory surface. The all-rate estimand prevents the argument from collapsing onto one fortunate learning rate, while five checkpoint stages reveal whether an

advantage emerges early, accumulates gradually, or disappears. The result remains specific to the tested DenseOn adaptation problem; it is stronger than a best-cell comparison and weaker than a universal optimizer ranking.

Geometry must earn explanatory status. Muon’s spectrum and NorMuon’s row adaptation describe how their updates are constructed. Our tests require more: geometry must predict corpus ranking at unseen doses beyond optimizer identity, stage, and dose. If no feature passes, the optimizer effect remains unexplained; the most visually distinctive statistic does not become a mechanism by default.

Dimension count and dimension utility are different. Random-removal robustness can arise from redundancy, distributed coding, or cancellation between helpful and degrading coordinates. Effective rank can also disagree with the functional ablation readout. The coordinate analysis therefore asks which parts of the representation help retrieval, how distributed that utility is, and whether any optimizer difference survives a change of basis.

Continuation tests state-dependent responses, not a gain decomposition. The crossed design asks whether a source-state advantage persists on average after further adaptation and whether the relative continuation preference changes between the fixed states. Neither an averaged main effect nor an interaction alone establishes benefit within both states, additivity, or full-trajectory co-adaptation. Explaining a particular geometric feature causally would require an intervention on that feature, beyond the present state-level comparison.

9 Conclusion

Across the four-rate surface, Muon versus AdamW is positive at +0.0064 [+0.0056, +0.0071] nDCG@10, and NorMuon versus AdamW is positive at +0.0154 [+0.0131, +0.0178].

Coordinate-level utility, its tested-basis sensitivity, and held-out-dose prediction are distinct results. Shared-rotation sign stability does not establish arbitrary-basis invariance or semantic independence of dimensions; these measurements alone do not identify a mediator of an optimizer effect.

After optimizer-state reset and calibration-probe scale matching, the crossed 50K continuation finds state, operator, and interaction contrasts of +0.0200 [+0.0100, +0.0300], +0.0200 [+0.0100, +0.0300],

and +0.0200 [+0.0100, +0.0300]. The relative Muon-versus-AdamW continuation effect is more favorable from the Muon-reached state at the tested horizon; this need not mean Muon helps from either state alongside a positive averaged source-state contrast and a positive averaged continuation contrast. This interaction does not establish full-trajectory co-adaptation.

The broader lesson is methodological: a retrieval optimizer should be explained by the functional weight states its closed-loop trajectory creates. Intrinsic properties of one update can motivate a hypothesis; state measurements tied to unseen retrieval and conditional continuation responses provide complementary evidence without constituting a causal decomposition of an empirical gain.

10 Limitations

Primary inference concerns one 149M-parameter DenseOn checkpoint, one English supervised corpus, and one epoch of full-parameter adaptation. The base model was pretrained with an Adam-family optimizer, so this is a study of optimizer switching rather than Muon-from-scratch pretraining. Four learning rates provide a broad but finite grid. The all-rate contrast reduces best-cell selection dependence but remains a one-seed, pinned-grid result; the secondary selected-recipe comparison does not replace replication across training views.

Decontamination reduces known benchmark overlap but cannot establish the absence of every semantic duplicate. Aggregate MTEB outputs lack per-query rankings, so uncertainty is computed over tasks and we make no query-level BEIR significance claim. The geometry and dimension bridges are predictive rather than causal and may miss useful properties outside the frozen candidates. Coordinate ablation uses an eight-document probe, and coordinate identities are not invariant under rotation; the rotation control limits, rather than eliminates, this dependence.

The crossed reset-state factorial concerns one fixed 60% checkpoint pair and a 50K-query continuation horizon. Its three order seeds do not replicate source-model training, and its marginal intervals are not family-wise controlled. It changes entire source states and continuation rules, not a measured geometry feature, and therefore does not identify mediation over the full 500K trajectory or establish why a feature predicts retrieval.

11 Ethical Considerations

This study does not collect new human-subject data or deploy a retrieval system to users. Released models and corpora can nevertheless contain social biases, harmful material, copyrighted text, or personal information. Higher benchmark nDCG is not evidence of fairness, safety, privacy, or deployment readiness. A production retriever requires corpus-specific auditing, filtering, access controls, and human review appropriate to its users and failure costs.

We report useful wall time, throughput, GPU-hours, and optimizer-state size because an optimizer recommendation has resource consequences. These measurements are specific to the pinned hardware and software stack and do not justify unbounded hyperparameter search. Public artifacts must respect the licenses and usage terms of upstream models, datasets, and evaluation collections.

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A Artifact and Reproducibility

We preserve the exact model and dataset revisions, deterministic 500K training ledger, negative sampling seeds, optimizer routing, package locks, W&B histories, checkpoint hashes, and evaluation receipts. Every reported table is generated only after source-bound completeness and provenance audits. The public artifact manifest binds each model state and result to its byte count and digest; the renderer fails closed when an expected run, checkpoint, task, or source identity differs.

B Complete Weight-Space and Dimension Readouts

The main text reports the weight- and representation-space findings that bear on the mechanism. For completeness, the tables

below retain every frozen candidate, including negative results.

C State-by-Operator Mechanism Test

The crossed experiment resets optimizer state at both source checkpoints, matches the fixed calibration-probe update scale, and reports all three co-primary contrasts regardless of sign. Exact seed-task contrasts and all five probe stages remain in the source-addressed artifact tables.

Geometry feature	RMSE reduction	Improved	Defined	Criterion	Residual ρ
log segment/weight norm	+0.000355457	3/4	4/4	supported	-0.809
segment stable-rank fraction	+1.18898e-07	2/4	4/4	not supported	-0.077
segment effective-rank fraction	+1.18898e-07	2/4	4/4	not supported	-0.077
segment row-norm CV	+1.18898e-07	2/4	4/4	not supported	-0.077
segment top-1% row energy	+1.18898e-07	2/4	4/4	not supported	-0.077
cumulative/weight norm	+6.63813e-05	3/4	4/4	supported	-0.393
cumulative stable-rank fraction	+1.63128e-06	3/4	4/4	supported	-0.064
segment subspace overlap to AdamW	-0.000335231	0/4	4/4	not supported	+0.139
cumulative subspace overlap to AdamW	-0.000335231	0/4	4/4	not supported	+0.139

Table 2: Every original weight-space predictor under four-fold leave-dose-index-out prediction. Decisions use exact MSE comparisons, not displayed RMSE. A missing correlation is undefined, not zero; baseline equivalence is distinct from an unidentified or numerically unresolved prediction.

Feature	Contrast	Effect	Simultaneous 95% CI	Decision
helpful mass share	Muon–AdamW	-0.00368479	[-0.0163172, +0.00894758]	inconclusive
helpful mass share	NorMuon–AdamW	+0.0039706	[-0.0117415, +0.0196827]	inconclusive
helpful mass share	NorMuon–Muon	+0.00765539	[-0.00720191, +0.0225127]	inconclusive
helpful participation	Muon–AdamW	-0.00407581	[-0.0135843, +0.00543271]	inconclusive
helpful participation	NorMuon–AdamW	-0.00167711	[-0.0122436, +0.00888938]	inconclusive
helpful participation	NorMuon–Muon	+0.0023987	[-0.0033673, +0.00816471]	inconclusive
degrading mass	Muon–AdamW	+0.000486392	[-0.00378336, +0.00475615]	inconclusive
degrading mass	NorMuon–AdamW	-0.000196003	[-0.00507519, +0.00468318]	inconclusive
degrading mass	NorMuon–Muon	-0.000682395	[-0.0053668, +0.00400201]	inconclusive

Table 3: All nine final-stage co-primary contrasts, averaging every declared rate. One common-resample max- T family covers all entries; task resampling does not estimate training-seed variation.

Feature	RMSE reduction	Improved	Defined	Criterion
helpful mass share	-7.30108e-06	1/4	4/4	not supported
helpful participation	-4.46922e-06	0/4	4/4	not supported
degrading mass	-1.51834e-05	0/4	4/4	not supported
50% random-removal retention	+2.06253e-07	2/4	4/4	not supported

Table 4: All four functional predictors beyond optimizer, stage and within-optimizer dose. Decisions use exact MSE comparisons; displayed RMSE is not a significance test. Undefined folds are retained, never pooled as available cases.

Factorial contrast	Effect	95% CI	Decision
Muon-state – AdamW-state	+0.0200	[+0.0100, +0.0300]	positive
Muon operator – AdamW operator	+0.0200	[+0.0100, +0.0300]	positive
State $\times$ operator interaction	+0.0200	[+0.0100, +0.0300]	positive

Table 5: Prospective reset-state factorial on final decontaminated-BEIR nDCG@10. Effects average the fixed three order seeds and 14 tasks; intervals use the frozen two-way seed/task cluster bootstrap. These are marginal 95% intervals, not simultaneous family-wise intervals.

D Claim–Evidence Firewall

E Full Experimental Matrix

The DenseOn matrix contains three optimizers, four rates per optimizer, five checkpoints, and 14 tasks: 12 runs, 60 checkpoints, and 840 checkpoint–task units. The released result tables additionally contain every optimizer-by-stage mean, validation-selected secondary contrast, per-task final score, wall time, throughput, peak allocation, and checkpoint size. No run is removed on the basis of retrieval quality.

Claim	Required evidence	Insufficient evidence alone
Faster adaptation	useful wall time, examples/s, GPU-hours, matched-hardware time-to-quality	lower loss at equal steps
Better DenseOn optimizer	all-rate contrast over 14 tasks	best BEIR cell alone
Weight motion explains retrieval	held-out prediction beyond optimizer, stage, and dose	larger displacement alone
Dimensions are used more constructively	helpful mass and participation rise while degrading mass falls, with simultaneous intervals	effective rank alone
Geometry predicts retrieval	held-out prediction beyond optimizer, stage, and dose	optimizer-identifying geometry
Source-state advantage persists on average after continuation	crossed endpoint state contrast with calibrated probe scale	one-step functional proxy
Continuation preference depends on reached state	crossed endpoint interaction	full-trajectory co-adaptation

Table 6: The minimum evidence required for each headline claim.